# A Multi-Agent AI System for Real-Time Detection and Classification of Dark Patterns on Live Webpages

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*Abstract*— **The user authentication mechanism in Microsoft Windows is a crucial element of contemporary digital security. For penetration testers, red teams, and digital forensic experts, breaching this security barrier is a critical goal. This paper provides a comprehensive summary of the methods and resources used to carry out a bypass of the Windows login. We explore the evolution of these methods, starting from traditional offline attacks aimed at the Security Account Manager (SAM) database to more sophisticated "live" attacks that take advantage of hardware interfaces and vulnerabilities in contemporary authentication systems such as Windows Hello. A key finding from this survey is that the complex, hardware- integrated security framework of current Windows 10 and 11 systems—including Secure Boot, the Trusted Platform Module (TPM), and Virtualization-Based Security (VBS)—has made many outdated bypass methods ineffective. As a result, the attack surface has shifted to specific, patchable vulnerabilities in live system components and protocols. This paper offers a classified summary of these attack methods, evaluates the effectiveness of relevant tools, and explores the difficulties and potential research directions, especially the growing dangers associated with AI- powered autonomous attack agents.**

***Keywords***— **Windows Security, Login Bypass, Red Teaming, Penetration Testing, SAM Database, Windows Hello, Credential Theft**

## I. INTRODUCTION

### *A. Background: Dark Patterns in Cybersecurity*

Over the last decade, digital services have increasingly relied on persuasive interface design to drive engagement, data collection, and monetization, blurring the line between legitimate nudging and outright manipulation. Dark patterns occupy the manipulative end of this spectrum: they are interface and interaction strategies that intentionally steer users toward decisions that favor the service provider while compromising user autonomy, privacy, or security. These patterns often appear subtle in isolation—such as a pre-selected checkbox, a confusing opt-out link, or a misleading button label—but their cumulative impact can dramatically shift user behavior away from informed and privacy-respecting choices.

In security and privacy contexts, dark patterns become especially problematic because they operate at critical decision points—account creation, permission granting, consent to data processing, and security configuration—where users already face complexity and asymmetries of information. A user who is rushed, distracted, or lacks technical expertise may accept invasive defaults, disable protective settings, or ignore risk warnings simply because the interface has been engineered to make the safer path less salient or more burdensome. Consequently, dark patterns can be seen as a human-facing extension of the attack surface: instead of exploiting software vulnerabilities, they exploit predictable weaknesses in human cognition and attention to weaken overall security posture.

### *B. Problem Statement and Motivation*

Initially, work on dark patterns focused on ethical questions in user experience design, highlighting how commercial pressures incentivize manipulative layouts and wording. More recent studies, however, document concrete harms, such as over-collection of personal data, difficulty in exercising rights (e.g., data access or deletion), and persistent friction when trying to unsubscribe or opt out of tracking. From a cybersecurity viewpoint, these harms translate into larger attack surfaces, longer data retention windows, and misconfigured security settings that expose users to threats well beyond a single interaction.

At the same time, regulators and standard-setting bodies have begun to recognize that formal legal compliance is insufficient when interfaces are deliberately engineered to undermine the spirit of consent and transparency requirements. Enforcement agencies report widespread deployment of deceptive design in cookie banners, mobile apps, and subscription services, yet they face practical barriers in identifying violations at scale, gathering robust evidence, and translating technical design choices into legal categories. On the technical side, security and privacy engineers lack mature, standardized tools to automatically inspect interfaces, classify manipulative patterns, and integrate such checks into development pipelines or browser-side protections.

These developments motivate a systematic review that connects conceptualizations of dark patterns with empirical evidence, detection methods, and regulatory discourse, specifically through the lens of cybersecurity and user privacy. The goal is not only to describe what dark patterns are, but to understand how they undermine secure behavior, why current defenses are insufficient, and which technical innovations could support more trustworthy digital infrastructures.

### *C. Research Objectives and Contributions*

Dark patterns do not exist in a vacuum; their form and impact depend heavily on the context of use. On the open web, studies of e-commerce, news, and service websites reveal widespread use of dark patterns in consent management, pricing displays, and checkout flows. In many cases, privacy-relevant decisions are bundled with other tasks, such as accessing content or completing a purchase, making it difficult for users to focus specifically on security implications. On mobile platforms, operating system-level permission dialogs and app store policies interact with app-specific interfaces, creating layered opportunities for manipulation—for example, apps that request broad permissions immediately after installation, or that misrepresent why a permission is needed.

Platform ecosystems, such as social media and large marketplaces, add another layer of complexity. Their sheer scale allows subtle design tweaks to affect millions of users, and their control over recommendation and ranking systems can indirectly penalize privacy-protective behavior by making it less visible or less rewarded. At the same time, these platforms are often subject to stricter regulatory scrutiny, prompting both more sophisticated dark-pattern strategies and more advanced compliance tooling.

An emerging line of work considers how dark patterns affect or even target autonomous agents and AI-driven tools that interact with the web on behalf of users. Experiments with web-navigating agents show that deceptive flows can mislead not only humans but also automated systems, causing them to accept unfavorable defaults or fail to complete privacy-preserving tasks when interfaces are adversarially designed. This raises new cybersecurity concerns: as users increasingly rely on agents to manage settings or perform routine tasks, service providers could attempt to optimize interfaces not only against human cognition but also against the algorithms that act as user proxies. Understanding these contexts reinforces the need for detection frameworks that can operate across devices, platforms, and both human and agent-mediated interactions.

## II. FOUNDATION OF DARK PATTERNS AND CYBERSECURITY

### *A. Definitions and Taxonomies of Dark Patterns*

Dark patterns are commonly defined as user interface and interaction designs that deliberately steer, pressure, or mislead users into making choices they would not have made under neutral and clearly presented conditions. What separates a dark pattern from ordinary poor design is intentionality: the pattern is crafted to systematically benefit the service provider—through more data, more time on site, or more revenue—while diminishing user autonomy, privacy, or security.

Over time, several taxonomies have been proposed to organize this design space. Early work introduced families such as obstruction, sneaking, interface interference, and forced action, each capturing a distinct manipulation strategy (for example, hiding opt-out options, disguising advertisements, or forcing additional steps for account deletion). Later research moved toward unified ontologies that reconcile overlapping categories from different taxonomies and explicitly link concrete pattern instances—such as confirmshaming, trick questions, pre-ticked checkboxes, or forced continuity—to higher-level concepts and manipulation goals. These ontologies provide a shared vocabulary that can be used across disciplines and are particularly important for building automated detectors, annotating datasets, and discussing dark patterns in regulatory and legal contexts.

### *B. Cognitive Biases and User Decision-Making*

The effectiveness of dark patterns is grounded in well-known cognitive biases and limitations that shape how people make decisions online. Users tend to:

1) Stick with default options, especially when alternatives require extra effort or are hidden in secondary interfaces.
2) Rely on quick heuristics when faced with time pressure, complex terminology, or very long privacy notices.
3) React more strongly to perceived losses than equivalent gains, a phenomenon that can be exploited through framing.

Designers can exploit these tendencies by making risky or intrusive choices the default, more visually prominent, or less cognitively demanding than safer alternatives. For instance, a pre-checked consent box combined with a short, reassuring label will often result in more data being collected than an explicit opt-in with clear explanations. Similarly, descriptions that emphasize what a user "loses" if they disable personalization or reject tracking can discourage privacy-preserving behavior even when the actual functionality impact is small.

In practice, users frequently engage in satisficing: they select any option that appears to complete the immediate task—gaining access to content, finishing a purchase, or installing an app—rather than optimizing for long-term security and privacy. Dark patterns deliberately align this "path of least resistance" with decisions that increase data collection, retention, and exposure, effectively embedding a form of continuous, low-level social engineering within routine user interfaces.

### *C. Security and Privacy Implications*

From a cybersecurity perspective, dark patterns effectively transform the user interface into a soft but significant component of the attack surface. When users are nudged into

accepting broad tracking, linking multiple identities, or turning off security safeguards, the system's overall risk profile changes—even if cryptographic mechanisms and back-end defenses remain strong.

Key implications include:

1) Expanded data footprint: Interfaces that default to maximum data sharing, combine consents, or discourage granular control lead to storage of more data, of higher sensitivity, and for longer periods. This amplifies the fallout of data breaches, insider misuse, and secondary repurposing of data.

2) Persistent accounts and dormant data: Dark patterns that complicate account deletion or make cancellation difficult keep dormant accounts and associated data active, prolonging exposure even when users believe they have left the service.

3) Weakened protective behaviors: If multi-factor authentication enrollment is framed as optional, annoying, or time-consuming, fewer users will adopt it. If permission prompts are frequent, intrusive, or confusing, users may habituate to clicking "Allow" without reflection.

These dynamics create a gap between formal compliance and effective protection. A platform may technically provide consent controls, privacy dashboards, or security toggles, yet implement them in ways that discourage real use. For security and privacy professionals, this means that assessing risk cannot stop at checking whether the "right" functions exist; it must also examine how those functions are presented and whether dark patterns undermine their intended purpose.

## III. TAXONOMY OF DARK PATTERNS RELEVANT TO CYBERSECURITY

### *A. Classification by Manipulation Goal (data, money, attention)*

For cybersecurity-focused analysis, it is useful to group dark patterns by the primary goal they pursue. Three broad manipulation goals are particularly relevant:

1) Maximizing data extraction
Patterns in this group are designed to increase the amount, sensitivity, or persistence of collected data. Typical strategies include bundled consents, pre-selected tracking, vague justifications for data collection, and "all-or-nothing" choices that force users into sharing more data than necessary. These patterns directly affect confidentiality by expanding the data an attacker could eventually access.

2) Strengthening lock-in and commitments
These patterns make it easy to sign up, link accounts, or authorize third-party integrations, but difficult to cancel subscriptions, revoke permissions, or delete profiles. They often involve multi-step cancellation flows, hidden options, or confusing wording that discourages disengagement. From a security perspective, this keeps credentials and personal data active longer than users intend, increasing the window during which they can be compromised.

3) Maximizing engagement and exposure
Here, designs aim to keep users engaged and connected, for instance by nudging them to stay logged in, enable persistent notifications, or skip logout on shared devices. While framed as convenience, these choices can weaken practices like locking screens, logging out from public terminals, or limiting notification content on lock screens.

Classifying dark patterns by manipulation goal helps directly link interface strategies to specific security properties—confidentiality (extra data), integrity (configuration and permissions), and availability of user control (difficulty in reversing decisions).

### *B. Classification by Technical Channel (UI, notifications, consent flows)*

Another dimension of the taxonomy considers the technical channel through which manipulation is implemented. Three levels are especially important:

1) Element and layout level
At this level, manipulation is embedded in the static structure of the page or screen. Examples include asymmetric button styling (large, colorful "Accept" vs. small, gray "Decline"), misleading labels, hidden controls behind expandable panels, and visual clutter around safer options. Detection here often relies on analyzing HTML structure, layout, and visual prominence.

2) Notification and timing level
Patterns at this level exploit when and how prompts appear. "Nagging" through repeated dialogs, timed pop-ups during moments of high engagement, or consent requests triggered when users are distracted can push users toward quick, permissive decisions. In mobile environments and IoT, background notifications and permission prompts can be orchestrated to appear in contexts where users are less likely to evaluate them carefully.

3) Flow and journey level
Some dark patterns only emerge when looking across multiple steps. For example, a one-click sign-up contrasted with a multi-step, confusing cancellation process, or a consent journey where key options are scattered across several screens and menus. Flow-level patterns often require analyzing interaction sequences and state transitions rather than single screens.

For cybersecurity, all three channels matter: element-level design affects what users see; timing affects when they are most vulnerable to manipulation; and flow-level structure determines how hard it is to adopt and maintain secure, privacy-protective configurations.

### C. Contexts: Web, Mobile, Platforms, and Agents

Dark patterns do not look the same in every environment. Their form and impact depend strongly on the context in which users interact:

1) Web applications and e-commerce
Studies of shopping sites and web services have documented widespread use of dark patterns in cookie banners, checkout flows, upselling, and account-management interfaces. Security-relevant decisions—like agreeing to data sharing or saving payment details—are often embedded in these flows, where users are focused on completing a purchase rather than on long-term risk.

2) Mobile apps and IoT devices
On mobile platforms, system-level permission dialogs interact with app-specific designs. Apps can request broad permissions at installation, repeatedly prompt until users relent, or frame permissions as mandatory for basic use. In home IoT, limited displays and constrained inputs are used to justify simplified, often more intrusive defaults, making it difficult for users to understand or change security and privacy settings.

3) Large platforms and ecosystems
Social networks, app stores, and marketplaces operate at a scale where subtle dark patterns can affect millions of users simultaneously. Interface decisions around recommendation settings, visibility of privacy controls, and default audience settings can materially influence how much personal and behavioral data is exposed.

4) Agents and AI-mediated interactions
Emerging work shows that dark patterns can also affect autonomous web agents and AI tools that act on behalf of users, causing them to choose unfavorable defaults or fail to identify protective options in adversarial interfaces. As more users rely on such agents to manage settings or perform routine tasks, service providers may start optimizing against both human and machine decision processes, adding a new layer of complexity to cybersecurity and usability.

Recognizing these contextual differences is essential when designing detection systems and countermeasures. A taxonomy that works for desktop web may need to be adapted for mobile, IoT, or agent-based scenarios, and security implications may vary accordingly.

## IV. DATASETS AND BENCHMARKS USED IN DARK-PATTERN RESEARCH

### A. Public Web Datasets and Crawls

A substantial portion of dark-pattern research relies on large-scale crawls of commercial websites and services to study how manipulative designs appear "in the wild." In these studies, researchers automatically visit thousands of webpages—typically e-commerce, news, and service sites—and capture HTML, CSS, and screenshots of key flows such as sign-up, cookie consent, checkout, and account-management interfaces. The collected pages are then manually or semi-automatically annotated to mark the presence of specific dark-pattern categories, creating corpora that support both prevalence analysis and training of detection models. For example, early work on shopping websites identified hundreds of distinct instances of obstruction, sneaking, and forced continuity across more than ten thousand sites, establishing that dark patterns are not isolated anomalies but systematically deployed at scale. Subsequent crawls extended this approach to consent banners after major privacy regulations, demonstrating how interface configurations converge on certain manipulative variants even under legal pressure.

### B. Manually Labeled Dark-Pattern Corpora

Beyond raw crawls, several studies have constructed

manually labeled corpora where expert coders or trained annotators systematically classify interface elements and flows according to a predefined taxonomy. In these datasets, each instance—such as a particular dialog, banner, or step in a flow—is tagged with detailed labels that capture both its superficial form (for example, pre-ticked checkbox, countdown timer, or misleading hierarchy) and its underlying dark-pattern category (such as obstruction, interface interference, or privacy deception). Some corpora also record contextual metadata, including the domain, platform type, and regulatory jurisdiction, enabling cross-cutting analyses of how patterns differ across sectors and regions. High-quality annotation is labor-intensive, often requiring iterative reconciliation between annotators to ensure consistent labeling, but it is essential for training and evaluating supervised learning models that aim to detect dark patterns automatically. These labeled datasets also serve as reference material for legal and policy discussions, providing concrete, documented examples that illustrate otherwise abstract taxonomic categories.

### C. for Agents and Automated Tools (e.g., DECEPTICON-style environments)

More recent work has begun to construct benchmarks specifically designed to evaluate automated tools and intelligent agents navigating dark-pattern-rich environments. In these settings, agents are tasked with completing realistic goals—such as subscribing without unwanted add-ons, configuring privacy-friendly settings, or uninstalling or deleting an account—while interacting with interfaces that may or may not contain dark patterns. The benchmarks typically include instrumented environments that log the agent's actions, the sequence of states encountered, and the presence or absence of known dark pattern instances, enabling quantitative evaluation of both detection and robustness. Some approaches integrate large language models or knowledge-driven rules to guide exploration, allowing agents to reason about natural-language text, visual cues, and multi-step flows when deciding which actions to take. These benchmarks are particularly relevant for future multi-agent systems, as they provide controlled settings where perception, reasoning, and action can be tested jointly, and where improvements can be measured against standardized tasks and metrics rather than ad-hoc examples.

## V. DETECTION METHODOLOGIES AND TOOLING

### *A. Rule-Based and Heuristic Approaches*

Early work on dark-pattern detection primarily relied on rule-based and heuristic approaches, where expert knowledge about manipulative designs was encoded directly into detection logic. In these systems, taxonomies of dark patterns were translated into concrete conditions over page structure, text, and simple visual properties, for example by flagging dialogs with pre-selected checkboxes for non-essential tracking or identifying links whose labels do not match their actual actions. This direct mapping from taxonomy to rule made such tools intuitive to understand, as each rule could be traced back to published definitions and examples.

A key advantage of rule-based and heuristic methods is their transparency. When a detector reports that a consent banner is problematic because the "Accept all" button is visually dominant while the "Reject" option is hidden in secondary text, the underlying rule can be inspected and discussed by designers, lawyers, and regulators. This interpretability is valuable in compliance and enforcement settings, where stakeholders need to understand not just that a dark pattern was detected, but why it qualifies as such and how the interface might be improved.

However, these approaches are also brittle and labor-intensive to maintain. As designers introduce new layouts, microcopy, and interaction patterns, static rules often fail to capture conceptually similar but superficially different manifestations of dark patterns. Rules tuned for specific domains or languages may not generalize well to other sectors or locales, and expanding coverage requires continuous manual updates. This limits the scalability of purely rule-based systems, especially as dark-pattern strategies evolve in response to regulatory scrutiny and public awareness.

### *B. Machine Learning and NLP/Computer Vision Models*

To address the limitations of hand-crafted rules, more recent research has explored machine learning approaches that learn to recognize dark patterns from annotated examples. One strand of work focuses on natural language processing, applying models to button labels, link texts, prompts, and explanatory copy to identify patterns of misleading wording, asymmetric framing, or emotional manipulation. These models can capture nuanced linguistic cues that are difficult to enumerate in rules, such as confirmshaming phrases or subtly misleading descriptions of data use.

Another strand leverages computer vision and layout analysis on screenshots to capture visual aspects of manipulation. Convolutional and transformer-based architectures are used to analyze color, size, position, and contrast, enabling detectors to assess which elements are visually salient and how choices are presented relative to each other. By combining structural information from the DOM with visual features from rendered pages, these methods can detect cases where, for example, an "Accept" button is made large and bright while a functionally equivalent "Decline" link is small and low-contrast.

Hybrid learning-based models combine textual, structural, and visual signals into a unified representation, enabling richer classification of both single screens and multi-step flows. Given sufficient training data, such models can generalize beyond the specific patterns and wordings seen during training, capturing the underlying manipulation strategies rather than only their most common instantiations. This adaptability makes them attractive for monitoring large and diverse sets of websites and apps, where manual rule maintenance would be impractical.

Despite these strengths, machine-learning approaches introduce their own challenges. They require large, high-quality labeled datasets that are representative of different domains, languages, and regulatory contexts, which are costly to build and maintain. Moreover, many models act as black boxes: they output a classification score without a clear explanation, which complicates their use in settings where transparency and accountability are essential. Current work therefore explores ways to add explanation layers, such as highlighting influential regions of text or interface elements, but striking the right balance between performance and interpretability remains an open problem.

### *C. Implications for Defensive Security*

Recognizing that neither pure rules nor pure learning offers a complete solution, several studies advocate hybrid and semi-automated workflows that integrate automated detection with human expertise. In these workflows, automated tools act as filters that scan large numbers of interfaces and flag those with a high likelihood of containing dark patterns, while human reviewers investigate the flagged cases in depth. This division of labor allows organizations to focus expert attention where it is most needed, without requiring exhaustive manual inspection of every interface.

In practical implementations, hybrid systems often use rule-based checks for clearly defined issues—such as pre-ticked consent boxes or multi-step cancellation paths—and machine-learning models for more subtle patterns in language and visual design. The outputs from these components are aggregated into reports that present a prioritized list of suspected dark patterns, accompanied by screenshots, DOM snippets, and short descriptions of the detected issues. Reviewers then validate the findings, correct false positives, and, where appropriate, refine or add labels that can be fed back into model training and rule refinement.

Over time, this feedback loop can significantly improve detection quality. As experts annotate new or borderline cases, the system's understanding of what constitutes a dark pattern in particular regulatory or organizational contexts becomes more nuanced. At the same time, having a semi-automated pipeline encourages more regular auditing,

since the initial screening effort is handled by tools rather than by manual inspection alone. This is particularly valuable for large platforms, regulators, and consumer organizations that must monitor many services and track changes over time.

Hybrid workflows also highlight the importance of usable tooling and standardized outputs. To be effective, detection systems must present their results in forms that non-technical stakeholders can understand and act upon, such as visual dashboards, clear textual summaries, and exportable evidence packs for investigative or legal processes. They must align with shared taxonomies so that findings can be compared across audits and jurisdictions, and they must integrate smoothly into existing development, compliance, and review processes. In this sense, detection methodologies are not only an algorithmic challenge but also a socio-technical one, requiring careful design of interfaces, reporting formats, and collaboration practices between engineers, designers, lawyers, and regulators.

## VI. LEGAL, ETHICAL AND REGUALTORY PERSPECTIVES

### A. *Regulatory Definitions and Enforcement (EU, OECD, etc.)*

1) Many data-protection and consumer-protection frameworks now explicitly treat dark patterns as practices that can undermine valid consent, transparency, and fair dealing, especially when they pressure users into accepting intrusive defaults or obscure safer choices.

2) Regulators have begun referencing dark patterns in official guidance and enforcement actions, but they still face challenges in translating high-level legal concepts like “freely given” or “unfair” into concrete, testable criteria that apply across diverse digital services and jurisdictions.

### B. *Interdisciplinary Methods for Evidence Collection*

1) Because dark patterns sit at the intersection of law, design, and human behavior, evidence collection often combines doctrinal legal analysis with HCI methods such as interface coding, screenshot archives, and user studies to document both the presence of patterns and their effects on user understanding and autonomy.

2) Cross-disciplinary collaborations have produced coding schemes and ontologies that map specific interface elements and flows to legal categories and design taxonomies, but the lack of fully standardized procedures still makes it difficult to compare results and aggregate evidence across different investigations and research projects.

### C. *Industry and Policy Reports on Deceptive Design*

1) Large-scale audits by regulators, consumer groups, and researchers have revealed recurring dark-pattern tactics in consent banners, shopping flows, mobile apps, and subscription management, prompting policy reports that call for clearer design requirements and more proactive enforcement.

2) In response, some organizations are adopting internal ethical-design frameworks, pattern libraries, and review processes that explicitly discourage dark patterns and promote user-centric alternatives, although uptake remains uneven and often depends on regulatory pressure, reputational risk, and organizational culture.

## VII. IDENTIFIED RESEARCH GAPS

### A. *Limitations of Current Taxonomies and Coverage*

Despite significant progress, existing taxonomies and ontologies of dark patterns remain fragmented and incomplete. Some frameworks are rooted primarily in practitioner experience and emphasize recognizable pattern names and visual examples, while others are derived from legal analysis or psychological theory and focus on abstract constructs such as autonomy, coercion, or bias exploitation. This diversity of perspectives is valuable, but it leads to overlapping, inconsistent categories and makes it difficult to combine results across studies or apply taxonomies directly in automated detectors. Many taxonomies also focus broadly on consumer interfaces without explicitly modeling security- and privacy-specific behaviors, such as multi-factor authentication enrollment, permission granting on mobile, or account-recovery flows. As a result, current categorizations do not always capture the full range of ways in which dark patterns can degrade cybersecurity, nor do they provide clear guidance on how to prioritize patterns according to their security impact.

### B. *Gaps in Automated Detection and Real-Time Monitoring*

While rule-based and machine-learning approaches have shown promise in detecting dark patterns from static snapshots or curated datasets, there is still a notable gap in systems capable of real-time, in-the-wild monitoring. Most existing tools operate offline: they analyze collected screenshots, DOM dumps, or curated app sets, and are not integrated into browsers, development pipelines, or continuous compliance processes. This limits their usefulness for protecting users during actual interaction or for alerting organizations when deployed interfaces drift toward more manipulative configurations. Real-time detection poses additional challenges, including performance constraints, handling of dynamic and personalized content, and the need to avoid collecting unnecessary user data during inspection. There is also a lack of end-to-end systems that not only flag potential dark patterns but also reason about user goals, legal context, and possible remediation actions, such as suggesting alternative flows or

providing on-the-fly explanations to users.

### C. *Lack of Labeled Datasets and Cross-Platform Studies*

Another major gap lies in the availability of representative, longitudinal datasets that capture how dark patterns evolve over time and across platforms. Many existing datasets concentrate on a single domain—such as shopping websites, consent banners, or mobile apps—and are collected at one point in time, making it difficult to study trends, regulatory effects, or design responses to enforcement actions. Under-representation of certain sectors (for example, health, education, and government services), regions, and user populations also limits the generalizability of findings and raises questions about fairness and equity. Moreover, the rapid pace of interface experimentation, including A/B testing and personalization, means that patterns observed in one snapshot may differ significantly from those a few months later. Building and maintaining longitudinal datasets that track the same services over time, annotate changes in design, and link those changes to regulatory or market events would provide a much stronger foundation for evaluating interventions and training robust detection models

## VIII. FUTURE DIRECTIONS AND PROPOSED MULTI-AGENT SOLUTION

### A. *Requirements for Robust Dark-Pattern Detection Systems*

Future dark-pattern detection systems must be capable of operating reliably across heterogeneous platforms, devices, and jurisdictions while remaining tightly aligned with established taxonomies and emerging regulatory expectations. They need to process multiple input modalities—markup structure, rendered layout, textual content, and interaction traces—because manipulation can be encoded in visual prominence, wording, timing, or in how steps are sequenced in a user journey. Beyond accurate classification, these systems must produce outputs that are interpretable for non-technical stakeholders: rather than simply flagging a screen as "problematic," they should highlight specific elements, explain which dark-pattern categories are implicated, and indicate how these elements could be redesigned to respect user autonomy and security. At the same time, monitoring and analysis must themselves be privacy-preserving and secure, collecting only the minimal information needed to evaluate an interface and ensuring that detection components do not introduce new attack vectors or data-retention risks. To be practically useful, such systems must integrate into existing organizational workflows—as checks in design tools, continuous-integration pipelines, and compliance dashboards, and as lightweight, client-side protections that can warn end users during real-time interaction with potentially manipulative interfaces.

### B. *Role of Multi-Agent Architectures and Advanced Models*

Multi-agent architectures provide a compelling blueprint for meeting these requirements, because they allow different capabilities to be distributed across specialized, cooperating components. In a multi-agent setup, one agent can focus on structural and visual analysis, rendering webpages or app screens and examining their layout, color, and relative prominence of controls to detect suspicious asymmetries. A second agent can specialize in natural-language understanding, parsing button labels, consent texts, error messages, and explanatory copy to identify misleading framing, bundled purposes, or emotional pressure. A third agent can be tasked with mapping the combined visual and textual cues onto a formal taxonomy of dark patterns and relevant legal concepts, producing structured, human-readable reports for auditors, designers, or regulators. Coordination mechanisms enable these agents to share intermediate findings, reconcile disagreements, and adapt their exploration strategies as they encounter multi-step flows, such as sign-up, checkout, cancellation, or account deletion. By combining learned models with explicit knowledge about patterns and regulations, such a system can move beyond static classification toward context-aware reasoning, for example differentiating between similar dialogs that have different implications depending on the task, user goal, or jurisdiction.

### C. *Alignment with SDGs and Ethical Design Principles*

Advances in detection and multi-agent architectures must be embedded within a broader commitment to ethical, human-centered, and sustainable digital design. Technical systems that expose dark patterns are only effective if organizations are willing and able to act on their findings, revising interfaces to prioritize user autonomy, security, and privacy over short-term gains from manipulative engagement. Aligning detection tools with ethical-design frameworks and internal governance processes—such as pattern libraries, design reviews, and impact assessments—can help shift incentives away from exploitative tactics and toward transparent, trustworthy interactions. At the societal level, positioning dark-pattern mitigation alongside digital-rights and responsible-innovation agendas underscores that resisting manipulative design is not just a compliance issue but a question of fairness, inclusion, and long-term resilience of digital infrastructure. In this perspective, a multi-agent detection system becomes one component in a larger ecosystem of regulation, organizational practice, and user empowerment, supporting a gradual transition from opaque, provider-centric interfaces to environments where users can meaningfully understand, question, and control how their data and decisions are shaped.

## IX. CONCLUSION

The body of work reviewed in this paper shows that dark patterns are no longer a marginal curiosity of user experience design but a structural feature of many contemporary digital services with direct consequences for cybersecurity and privacy. Across

web, mobile, IoT, and large platform ecosystems, interfaces are routinely shaped to exploit cognitive biases, defaults, and information asymmetries, nudging users toward excessive data disclosure, weakened security configurations, and long-lived accounts and permissions that expand the overall attack surface. Existing taxonomies, ontologies, and empirical studies provide a detailed vocabulary for describing these practices and demonstrate their prevalence and impact, yet they also reveal fragmentation in concepts, gaps in coverage for security-specific behaviors, and significant under-representation of certain domains and user groups. Methodological advances in rule-based detection, machine learning, and hybrid auditing workflows offer promising tools for identifying dark patterns at scale, but most current systems operate offline, are limited to specific contexts, or struggle to deliver explanations that are robust enough for legal and regulatory use.

These limitations point to the need for a new generation of detection frameworks that are multi-modal, interpretable, and tightly integrated with both organizational processes and emerging regulatory expectations. In particular, framing the interface as a security-relevant surface suggests that dark-pattern mitigation should stand alongside more traditional technical controls as part of defense in depth, rather than being treated purely as an ethical or usability concern. The proposed direction of a multi-agent AI system for real-time detection and classification of dark patterns on live webpages responds directly to this need, distributing crawling, perception, classification, and explanation across cooperating agents that can reason jointly about structure, language, and user flows. When aligned with ethical-design frameworks and embedded into development, auditing, and user-facing tools, such systems have the potential to shift incentives away from manipulative design and toward transparent, trustworthy interactions. In doing so, they can help close the gap between formal compliance and substantive protection, supporting regulators, organizations, and end users in recognizing, contesting, and ultimately reducing the use of dark patterns in security- and privacy-critical digital environments.

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